

MANUFACTURING METHODS AND MATERIAL EMBODIMENTS

In various embodiments, the intraluminal compliance and distance profiling device may be manufactured using techniques selected to achieve durability, comfort, sealing integrity, and scalable production. Manufacturing approaches may vary depending on device tier, regulatory classification, and intended deployment environment while maintaining functional equivalence across implementations.

The insertable measurement section may be produced using overmolding processes in which compliant outer materials are molded around internal structural backbones and embedded sensing components. Overmolding may provide improved sealing, user comfort, and protection of sensitive electronics while maintaining precise alignment between sensing regions.

Suitable backbone structures may include flexible polymers, reinforced elastomeric cores, segmented articulated supports, or semi-rigid internal frames designed to balance structural stability with controlled flexibility. Backbone configurations may be selected to maintain sensor alignment during insertion while minimizing localized pressure concentration.

In certain embodiments, inflatable compliance regions may be manufactured using bonded elastomeric membranes, welded polymer films, molded silicone chambers, or layered composite structures configured to withstand repeated expansion cycles. Bonding methods may include thermal welding, adhesive bonding, chemical bonding, insert molding, or integrated co-molding techniques.

Sensing components may be integrated using flexible printed circuit boards, rigid-flex electronics assemblies, encapsulated sensor modules, or distributed microelectronic packages embedded within compliant materials. Encapsulation techniques may include conformal coatings, potting compounds, or hermetic sealing structures configured to protect electronics from fluid exposure.

Distal sensing windows may be manufactured from optically transmissive or acoustically transmissive materials selected to preserve measurement performance while protecting underlying sensors. Window structures may be flush-mounted, recessed, replaceable, or integrated into molded tip geometries.

The handle portion may be produced using injection molding, additive manufacturing, machining, or hybrid fabrication methods. Internal cavities may support placement of batteries, pumps, processors, communication modules, and vibration or feedback actuators while maintaining ergonomic handling characteristics.

Assembly processes may include modular subassembly construction in which insertable sections, inflatable components, and electronic control modules are manufactured independently and later combined into a unified device. Modular assembly may simplify manufacturing scalability, maintenance, and future product upgrades.

In certain embodiments, manufacturing tolerances and quality-control procedures may include automated testing of pressure integrity, sensor calibration verification, inflation response testing, communication validation, and electrical safety checks prior to distribution.

Accordingly, the disclosed manufacturing and material embodiments enable production of devices ranging from simplified consumer configurations to advanced research or clinical-support instruments

while maintaining consistent sensing performance, safety characteristics, and long-term durability.